

Data Dictionary

This document contains three tables that describe the variables used in the analysis, including definitions, units, and notes on transformations. The first table holds the core variables used in the analysis. The second table has the variables used to compute outliers using the IQR method. The third table contains the variables as they are written in the initial dataset.

1. Core Variables

Variable	Type	Description	Notes
Death_rate	Numeric	Mortality rate per 100,000 population for a specific cause, region, sex, age group, and year.	Converted from Value Numeric. Used for all statistical tests and plots.
Cause	Factor	Specific cause of death as defined by WHO (50+ categories).	Later grouped into broader categories.
Cause_group	Factor	Aggregated cause categories (e.g., Cardiovascular, Cancer, Respiratory).	Created manually using case_when () to simplify analysis.
sdg_region	Factor	WHO Sustainable Development Goal (SDG) region classification.	Used for regional comparisons and Chi-square test.
sex	Factor	Sex category: Male, Female, Both sexes.	Used for subgroup analysis and interaction plots.
Age_group	Factor	Age categories for older adults (e.g., 60-64, 65-69, 70-74).	Used for age-based comparisons and Kruskal–Wallis test.
year	Numeric	Year of observation (2000-2007).	

2. Data Cleaning

Variable	Type	Description	Notes
Q1	Numeric	First quartile of death_rate within each group.	Used for outlier detection.
Q3	Numeric	Third quartile of death_rate within each group.	Used for outlier detection.
IQR	Numeric	Interquartile range (Q3 - Q1).	Used to compute outlier thresholds.
lower	Numeric	Lower bound for outlier removal ($Q1 - 1.5 * IQR$).	Observations below this are removed.
upper	Numeric	Upper bound for outlier removal ($Q3 + 1.5 * IQR$).	Observations above this are removed.

3. Metadata Variables (from WHO dataset)

Variable	Type	Description	Notes
Value Numeric	Numeric	Raw numeric mortality value from WHO.	Converted to death_rate.
Age group	Character	Original age group label from WHO.	Cleaned into age_group.
SDG region	Character	Original region label from WHO.	Cleaned into sdg_region.
Sex	Character	Original sex label from WHO.	Cleaned into sex.
Cause	Character	Original cause label from WHO.	Cleaned into Cause.